

```

from myhdl import *

ACTIVE = 0
DirType = enum('RIGHT', 'LEFT')

def jcd2(goleft, goRight, stop, clk, q):
    """ A bi-directional 4-bit Johnson counter with stop control.

    I/O pins:
    =====
    clk      : input free-running slow clock
    goleft   : input signal to shift left (active-low switch)
    goRight  : input signal to shift right (active-low switch)
    stop     : input signal to stop counting (active-low switch)
    q        : 4-bit counter output (active-low LEDs; q[0] is right-most)

    """
    dir = Signal(DirType.LEFT)
    run = Signal(False)

    @always(clk.posedge)
    def logic():
        # direction
        if goRight == ACTIVE:
            dir.next = DirType.RIGHT
            run.next = True
        elif goleft == ACTIVE:
            dir.next = DirType.LEFT
            run.next = True
        # stop
        if stop == ACTIVE:
            run.next = False

```

A sample MyHDL code

to a hardware platform, testing is not that common. In case of Verilog designs, we can also use MyHDL as hardware verification language by co-simulation with traditional HDL simulators.

How MyHDL benefits users

In this section let us try to analyse some of the reasons why a hardware designer should consider MyHDL tool.

For beginners

Free and open source. As a beginner, you will be least interested in purchasing a costly software. MyHDL is a well-documented, open source software that a beginner can really benefit from.

Simplicity of Python. An engineer trying to master digital hardware design needs to put up a lot of hard work. Choice of a right developing language makes his or her task easier. Among high-level programming languages, Python is one of the

simplest to learn, and MyHDL aims to bring the same level of simplicity into hardware design.

Apart from this, the automated conversion of design into Verilog/VHDL also eases the job of a designer as a beginner.

For professionals

Latest software developments readily available for the designer. Many developments are going on in software methodologies which lead to major improvements in the

development process. These are emerging in communities of modern software languages, like Python.

As MyHDL is based on Python language, these developments are directly available to the hardware designer. An example is that of unit testing, previously mentioned in this article.

Use of dynamic language directly for hardware design. Dynamic languages, like Python, Perl and Tcl, play an important role in modern digital designs. Many developers prefer these languages to HDL languages, because they can get things done quickly with these. Again, MyHDL has significant advantage over other HDL languages as it uses a dynamic language (Python) directly to design hardware.

Unified algorithm and implementation. Usually, development of an algorithm and implementation of a design are done by different engineers in various platforms. En-

gineers for algorithm development commonly use Python, while HDL implementations are done by a different set of engineers in another platform.

MyHDL essentially unifies these two domains. Both can be developed in the same environment. Even though conversion from algorithm to an HDL implementation requires knowledge of HDL based designs, being able to do this in the powerful Python environment significantly increases the quality and productivity of the design. The designer can directly reuse the verification work in both domains, thereby increasing productivity.

A hardware design tool to experiment with ideas

We have already seen some merits of this wonderful software. The use of a high-level scripting language directly into hardware design offers many advantages over a conventional HDL language, such as improvement in speed and productivity. MyHDL cannot completely substitute the use of other HDL languages. For example, a designer wanting to have accurate timing simulations needs to go for industry standards, VHDL and Verilog.

Jan Decaluwe, the main developer of the software, envisions MyHDL as the hardware design tool of choice to experiment with new ideas. He strongly believes that hardware engineers can enjoy the benefits of a well-designed, widely-used language and the open source model behind it, giving the freedom to experiment with ideas. ●

The author is an electronics enthusiast from Kerala



For More
Test and Equipment Stories
Visit www.electronicstest.com